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Seasonal Changes in Heat Balance of Dogs Acclimatized to Outdoor Climate

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Abstract Heat production (M), dry heat loss ($R+C$), evaporative heat loss (E) and rectal temperature (T_{re}) were measured in a direct calorimeter in female mongrel dogs acclimatized to outdoor climate at Kanazawa (latitude; 36°35′ N), Japan. M and total dry and evaporative heat losses (H_L) were minimum at calorimeter wall temperatures (T_w) of 26–29°C in summer and 22–26°C in winter (thermoneutral temperature; TNT). The seasonal shift of the lower critical temperature was confirmed. At T_w below TNT, the values of M and H_L were significantly higher in summer. At T_w above TNT, these values increased. At TNT and above, M and H_L were significantly higher in winter. $(R+C)$ decreased linearly with increasing T_w in both seasons. At T_w below 26°C, $(R+C)$ were significantly higher in summer. At T_w above 26°C, E increased greatly. The values of E were significantly higher in winter at T_w 29–32°C. T_{re} remained nearly constant at TNT and below, and increased at T_w above TNT in both seasons. Mean body surface temperature ($\bar{T}_{sf}$) decreased with decreasing T_w. Body thermal conductance (K) was minimum at T_w below 26°C in summer and at T_w below 22°C in winter. At T_w above these temperatures, K increased significantly. Whole body insulation (I) was significantly higher in winter, particularly at T_w 18°C. These results suggest that the dogs reared outdoors in winter acclimatized to cold in two ways; by increasing the insulating effect of the fur coat and by elevating resting heat production.

Many mammals increase their pelage in the winter season against the cold and increase the metabolic rate to meet the strong and growing thermal demands of their environments (HART, 1956, 1961; HÉROUX, 1963; HÉROUX *et al.*, 1959; IRVING, 1964; IRVING *et al.*, 1955: SCHOLANDER *et al.*, 1950). Domestic dogs, commonly used as experimental animals in physiology, also increase the amount of their pelage when reared outdoors in winter. The lowering of critical temperature and the reduction of the metabolic response to cold are surmised in winter dogs. However, DURER and HANNON (1962) suggested from studies on calorie intake that

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cold-acclimated well-insulated Eskimo dogs elevate their metabolic rate over a wide range of temperature. NAGASAKA and CARLSON (1965) observed an increased metabolic rate at a thermoneutral temperature of 28°C in cold-acclimated dogs, the hair density of which was evidently increased. The effects of seasonal acclimation on the metabolic rate in dogs seem to be a matter of debate. Although HAMMEL *et al.* (1958) measured the changes in heat balance of unacclimated dogs using a direct calorimeter, a complete partitional calorimetric study on dogs acclimated to outdoor climate has not been reported. The present study was undertaken to determine the changes in heat balance by using a direct calorimeter in dogs reared outdoors and to contribute to the knowledge on the seasonal acclimation in this particular species.

METHODS

Three female mongrel dogs, born and raised in our laboratory, were studied in summer (August–September) and winter (January–February) in Kanazawa, Japan (latitude, 36°35′ N; altitude, 58 m). The dogs were maintained individually outdoors under a roof and provided with commercial dog food (Hokuriku Labo., Co., Toyama) and water *ad libitum*. The food consumption was determined every day. Body weights were determined weekly. The outdoor temperatures and humidity during the experimental periods and the characteristics of the animals are shown in Tables 1 and 2.

All measurements were made between 800 and 1,900 hr. The dogs were trained to sit quietly on a dog restrainer made of metal frames. At the time of measurement the dog was enclosed in the direct calorimeter, sitting quietly on the

Table 1. Outdoor temperatures and mean relative humidity.

	Summer	Winter
Mean temperature (°C)	25.3±0.3	3.1±0.4
Maximum temperature (°C)	29.7±0.4	6.7±0.5
Minimum temperature (°C)	21.7±0.3	0.1±0.3
Mean humidity (%)	74.8±1.0	71.9±1.2

Values are means±S. E.

Table 2. Body weight and food intake.

Dog	No. 1	No. 2	No. 3
Summer			
Body weight (kg)	7.99±0.06	9.30±1.10	8.27±0.05
Food intake (g/day)	262±13	281±15	278±8
Winter			
Body weight (kg)	8.20±0.03	9.81±0.06	9.00±0.06
Food intake (g/day)	412±11	429±13	422±8

Values are means±S.E.

restrainer. Dry heat loss ($R+C$) and evaporative heat loss (E), heat production (M) and rectal temperature (T_{re}) were continuously measured by the direct calorimeter (a modified Thermonetics SEC-A calorimeter, Thermonetics, San Diego, Cal.). The calorimeter wall temperatures (T_w) were kept at 18, 22, 26, 29 and 32°C with an accuracy of $\pm$0.02°C. Feces and urine were collected in a metal pan filled with vegetable oil placed underneath the dog restrainer to prevent unwarranted evaporation from the waste. The ($R+C$) was estimated from the output voltage and the calibration constant of the calorimeter. The air temperature in the climatic chamber (Built-Lab., Tabai MFG, Co., Osaka) in which the calorimeter was placed was kept at the same temperature as T_w (within$\pm$0.2°C). The expired air together with the air inside the calorimeter was withdrawn to the outside of the calorimeter with a pump at a rate of 20 liters·min^{-1} through the plastic hood loosely covering the dog's head. The same amount of fresh air was introduced into the calorimeter from the climatic chamber. The relative humidity of the air exhausted from the calorimeter was continuously monitored with a sensitive hygrometer (HM-1, Vaisala, Helsinki) and E was calculated from the relative humidity and the air flow. The mean of the relative humidity of the air inside the climatic chamber measured prior to and after the measurement was taken as the reference humidity of the air sent to the calorimeter by pump. The air flow was measured with a gas meter (Aichi-Dokei, Nagoya). A small fraction of the air (400 ml·min^{-1}) was collected in an anesthesia rubber bag (2 liters) every 5 min and the oxygen content of the exhausted air was measured with a Beckman E_2 oxygen analyzer. Oxygen consumption ($\dot{V}_{O_2}$) was calculated from the oxygen content and the flow of the air. Heat production was calculated from the $\dot{V}_{O_2}$ using the calorific equivalent of 4.8 kcal·liter^{-1} (20.1 kJ·liter^{-1}) (DAWSON *et al.*, 1970). The change in heat content of the air was estimated by measuring the temperature difference of the air leaving and entering the calorimeter. It was approximately 1% of the total ($R+C$) in this study. One hour after thermal equilibrium was attained for each dog (T_{re} and ($R+C$) became constant), all measurements were made for another 2 hr. Two measurements were made on the same day, one from 800 hr and another from 1,400 hr for another dog at the same T_w. For each T_w level, a total of 6 measurements was averaged.

Immediately after the measurement, the dog was replaced by a metal cylinder water container made to have the same surface area as that of the dog [SA (m^2)= 0.11$\times$body weight (kg$^{0.67}$)] (GUYTON, 1963). The water container was filled with water kept warm by an electric heater and agitated vigorously *via* an internal screw. The surface temperatures and ($R+C$) of the container were continuously recorded on the potentiometer. When the value of ($R+C$) became exactly the same as that obtained from the dog, the surface temperature of the container should have been identical to the mean surface temperature ($\bar{T}_{sf}$) of the animal since the surface temperatures measured at any point of the container must have been the same. Body thermal conductance [$K=(R+C)/(T_{re}-\bar{T}_{sf})$] and whole body insulation [$I=$

$(T_{re}-\bar{T}_{sf})/(R+C)$] of the dog were calculated.

The data were compared for detecting any statistical significance of changes in these parameters by the Student's t-test and by analysis of variance described by DUNCAN *et al.* (1977).

RESULTS

Figure 1 shows the resting heat production (M) at various calorimeter wall temperatures (T_w) in the summer and the winter dogs. In the summer dogs, M was lowest at T_w between 26 and 29°C. At T_w below 26°C, M increased linearly with decreasing T_w at a rate of 2.06 $W \cdot m^{-2} \cdot °C^{-1}$. At T_w 32°C, M was significantly higher than that observed at T_w 29°C. In the winter dogs, M was lowest at T_w 22 and 26°C, above which it increased with increasing T_w. At T_w 18°C, M was higher than that observed at T_w 22°C. At T_w above 26°C, M were significantly higher ($p<0.01$) in the winter dogs than in the summer dogs. On the contrary, at T_w below 22°C, M were lower in the winter dogs than in the summer dogs.

Figure 2 shows the dry heat loss ($R+C$) and evaporative heat loss (E) at T_w between 18 and 32°C in the summer and the winter dogs. ($R+C$), lowest at T_w 32°C, increased with decreasing T_w in both groups. At T_w between 26 and 32°C,

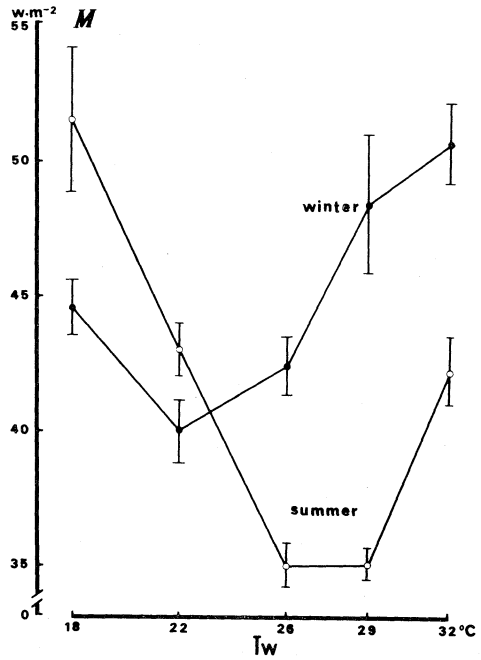


Fig. 1. Resting heat production (M) plotted against calorimeter wall temperature (T_w) in summer (○) and winter (●). Values are means±S.E.

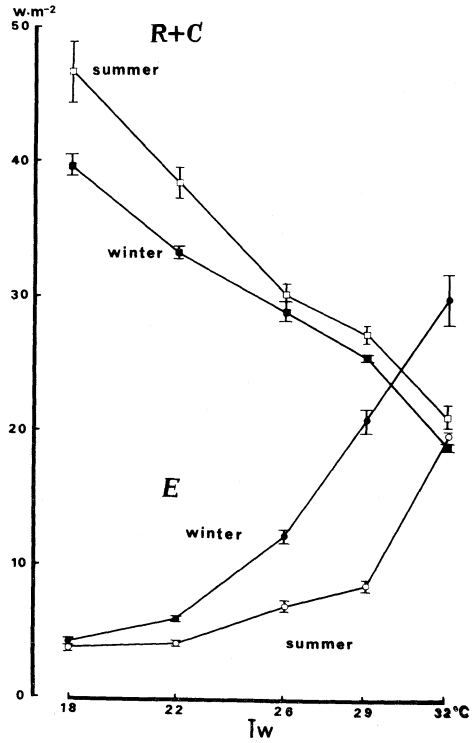


Fig. 2. Dry heat loss ($R+C$) and evaporative heat loss (E) plotted against T_w in summer (□, ○) and winter (■, ●). Values are means±S.E.

there was no significant difference between the ($R+C$) in the summer and the winter animals. At T_w below 26°C, ($R+C$) in the summer dogs was significantly higher ($p<0.01$) than that in the winter dogs. Evaporative heat loss (E) maintained at a low level at T_w 18 and 22°C, increased with increasing T_w above 26°C in both groups. E was significantly higher in the winter dogs, particularly at T_w 29–32°C ($p<0.01$).

The changes in the total heat losses (H_L) are shown in Fig. 3. In the summer dogs, H_L was lowest at T_w between 29 and 26°C, below which it increased linearly with decreasing T_w at a rate of 1.81 $W \cdot m^{-2} \cdot °C^{-1}$. At T_w above 29°C, H_L also increased. In the winter dogs, H_L was lowest at T_w 22°C, above which H_L increased with increasing T_w. At T_w above 26°C, H_L were significantly higher in winter dogs ($p<0.01$). H_L were significantly lower in the winter dogs than in the summer animals at T_w between 18 and 22°C ($p<0.01$).

Rectal temperature (T_{re}) remained nearly constant at T_w from 18 to 26°C in the summer dogs (Table 3). At T_w above 26°C, T_{re} increased significantly ($p<0.01$). T_{re} in the winter dogs remained nearly constant at T_w 18 and 22°C, above which it increased significantly in the winter dogs ($p<0.01$). Except at T_w 32°C,

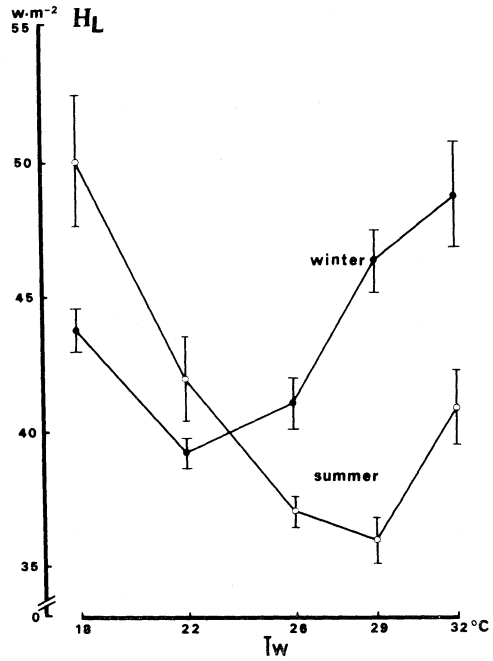


Fig. 3. Total heat losses (H_L), evaporative and nonevaporative, plotted against T_w. Values and symbols are the same as in Fig. 1.

Table 3. Rectal temperature and whole body conductance.

T_w(°C)	18	22	26	29	32
Summer					
T_{re}(°C)	37.85±0.10	37.90±0.08	38.00±0.05	38.30±0.04	38.56±0.06
K(W·m^{-2}·°C^{-1})	3.19±0.20	3.76±0.14	4.07±0.10	4.93±0.22	5.25±0.25
Winter					
T_{re}(°C)	38.03±0.08	38.03±0.04	38.27±0.05	38.48±0.03	38.54±0.03
K(W·m^{-2}·°C^{-1})	2.55±0.06	2.91±0.05	3.61±0.10	4.45±0.09	4.90±0.07

Values are means±S.E.

T_{re} in the winter dogs was always higher than that in the summer dogs. The difference between T_{re} in the summer dogs and the winter dogs was significant at T_w 26 and 29°C. The mean surface temperature ($\bar{T}_{sf}$) of the fur coat calculated from the method described above and the whole body insulation (I) in the summer and the winter dogs are shown in Fig. 4. $\bar{T}_{sf}$ decreased with decreasing T_w in both groups. There was no statistically significant difference between $\bar{T}_{sf}$ in the summer and the winter dogs at T_w between 26 and 32°C. At T_w below 22°C, $\bar{T}_{sf}$ became significantly lower ($p<0.05$) in the winter dogs than in the summer animals. The whole body conductance (K) decreased significantly at T_w 18 and 22°C in both groups (Table 3). There was no statistical significance between K in the summer

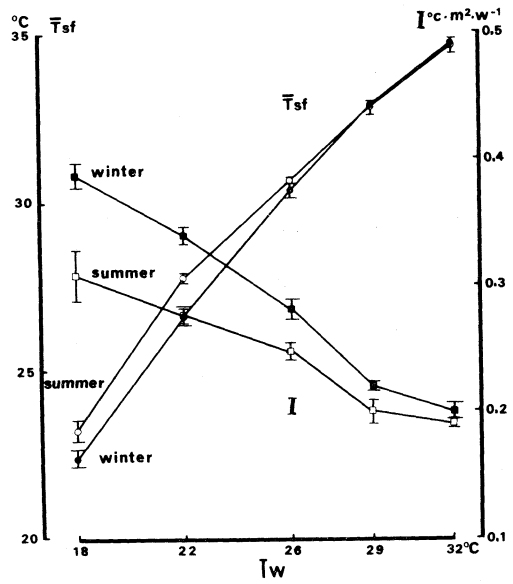


Fig. 4. Mean body surface temperature ($\bar{T}_{sf}$) and whole body insulation of the body (I) plotted against T_w. Values and symbols are the same as in Fig. 2.

and the winter dogs at T_w between 26 and 32°C. K was significantly lower in winter than in summer at T_w 18 and 22°C.

DISCUSSION

Rectal temperature (T_{re}) in the summer and the winter dogs was maintained at nearly the same level at T_w between 18 and 26°C (Table 3). At T_w 29 and 32°C, T_{re} increased significantly and a persistent increase in heat storage was suggested in both animals. As observed in Fig. 1, the lowest values of resting heat production (M) were recorded at T_w between 26 and 29°C in the summer dogs and at T_w between 22 and 26°C in the winter dogs. These results suggest that the critical T_w for metabolic increase shifted from 26°C in summer to 22°C in winter. The total dry and evaporative heat losses (H_L) changed in parallel to M in both groups (Fig. 3). At T_w above 29°C in summer and 26°C in winter, both M and H_L increased (Figs. 1 and 3). This evidence, together with some observations in literature (HAMMEL *et al.*, 1958; TAYLOR *et al.*, 1971), suggest that domestic dogs have a rather narrow thermoneutral zone in either season. Measuring heat balance with a direct calorimeter, HAMMEL *et al.* (1958) estimated that the lower critical temperature of domestic dogs was around 23 and 25°C. A similar result was obtained by TAYLOR *et al.* (1971) who compared changes in heat balance in dogs with African hunting dogs (*Lcaon pictus*). The African hunting dogs showed a much higher upper criti-

cal temperature and a wider thermoneutral zone. There also seems to be a great difference in the thermoregulatory profile of animals belonging to similar species. The upper critical temperature for metabolic increase seemed to be far lower in dogs, particularly in winter. Although the air was withdrawn from the calorimeter at a rate of 20 liters/min, apparent air movement was not noticed inside the calorimeter. Whether similar results can be obtained in dogs outside the calorimeter warrants further investigation.

At T_w from 18 to 26°C, heat balance was well maintained in both groups. At T_w 29 and 32°C in the winter dogs and 32°C in the summer dogs, M was apparently higher than H_L and a positive shift of heat balance was confirmed. As shown in Fig. 2, $(R+C)$ decreased linearly with increasing T_w in both groups. As reported by others (CHAPPUIS *et al.*, 1976), dry heat loss seemed to be passive and dependent on the ambient temperature. Thus, vasomotor activity in the body surface may not play an important role in regulating heat dissipation in the dog. Below the critical temperature, $(R+C)$ was significantly higher in the summer dogs (Fig. 2). It is difficult to understand that at low ambient temperature the cutaneous vessels dilate more strongly in the summer dogs than in their winter counterparts. The difference between the dry heat loss in both groups may be attributed to the seasonal difference in the thermal insulation of the pelage. According to HART (1956), the insulation of the fur shows a marked seasonal change in mammals weighing more than 1 kg. It is known that winter pelage increases in domestic dogs reared outdoors in winter. Dry heat loss is proportional to the areas of the body surface (HAMMEL *et al.*, 1958). However, the significant difference in $(R+C)$ cannot be explained by the difference in the body surface, because a marked postural change was not allowed during the measurement and it was hardly probable that the summer dogs extended their body surface more than the winter animals in cool environments.

Thermal conductance $[K=(R+C)/(T_{re}-\bar{T}_{sf})]$ increased with increasing T_w from 18°C (Table 3). However, the rate of increase in K was less at T_w between 18 and 26°C in the summer dogs and at T_w between 18 and 22°C in the winter dogs. At above critical temperature, K increased significantly in both groups. These results suggest that cutaneous vasoconstriction is close to maximum at the lower critical temperature so that a further decrease in cutaneous blood flow does not occur in the cold environment. Below critical temperature, vasomotor regulation is of minor importance in dogs and increased heat production plays a dominant role in thermoregulation, as in the case of monkeys (NAKAYAMA *et al.*, 1971; STITT and HARDY, 1971). Thermal conductance of dogs reported by HAMMEL *et al.* (1958) was between 7 and 12 $W \cdot m^{-2} \cdot °C^{-1}$ for all temperatures between 8 and 30°C, and considerably higher than the values obtained in this study. The discrepancy between the two results cannot be explained at present. Instead of using the mean skin temperature ($\bar{T}_s$), in this study, the mean surface temperature on the fur coat ($\bar{T}_{sf}$) was used. At ambient temperatures below body temperature, skin

temperature should always be higher than the surface temperature of the fur coat. The lower $\bar{T}_{sf}$ may be responsible for the considerably low values of K obtained in this experiment. Thermal gradient between the skin surface and the surface of the fur coat should be increased when the depth of the coat is increased. The lower $\bar{T}_{sf}$ and K, and the higher whole body insulation (I) of winter dogs particularly at lower ambient temperature may be attributed to the increased thickness of the fur layer. Hairs at the sites of measurement should be removed to measure skin temperature. Since the temperature of the bared skin is greatly influenced by the ambient temperature, the skin temperature thus recorded is not expected to be the same as that on the site which had been underneath the fur coat. It is practically impossible to measure the temperature on the pelage and to calculate the mean body surface temperature. The direct calorimetric method described here may be the most reliable one to obtain the mean surface temperature and to measure heat balance of furred animals.

Evaporative heat loss (E) increased greatly at T_w 32°C in summer and at T_w 29 and 32°C in winter (Fig. 3). To maintain the thermal balance in warm environments, the dog seems to depend largely on the increase in respiratory evaporative heat loss by panting (HAMMEL *et al.*, 1958; TAYLOR *et al.*, 1971). AOKI and WADA (1951) and IWABUCHI (1967) observed the existence of numerous histologically identifiable sweat glands in the hairy skin of the dog. However, the increased cutaneous evaporation could not be observed even at ambient temperature above 35°C (HAMMEL *et al.*, 1958). The blood circulating in the skin reduces body temperature effectively when the blood is cooled by sweat evaporation at high ambient temperatures. The higher cutaneous blood flow, if any, may play only a very minor role in reducing body temperature in the dog. At higher ambient temperatures, E was significantly higher in the winter dogs than the summer ones. Contrary to the cold environments, increased insulation of fur is not beneficial to maintaining heat balance in warm environments.

Below the critical temperature, resting heat production was significantly less in winter. This result seems to support the observations made by HÉROUX *et al.* (1959) on rats and by WEBSTER *et al.* (1969) on sheep. In contrast to rats conditioned to cold in the laboratory, the winter rats had a lower heat production than the summer rats over a considerable range of temperature (HÉROUX, 1959). These rats had a greater pelage insulation and lower skin temperature on the back and tail than the summer rats. NAGASAKA and CARLSON (1965) observed a consistently higher oxygen consumption of the cold acclimated beagle dogs at ambient temperature of 28 and 5°C. The effects of acclimation to constant cold in a laboratory may be different in the context of thermoregulation from the results of acclimation to the outdoor winter climate. The minimum resting heat production at the thermoneutral temperatures was significantly higher in winter dogs ($p<0.01$). As shown in Table 1, food intake increased greatly in winter without any significant increase in body weight. Similar seasonal changes in calorie intake were observed by

DURER and HANNON (1962), who observed that cold-acclimated dogs have an elevated metabolic rate over a wide range of temperature. In summary, the present study indicates that the seasonal variations of resting heat production as well as heat loss occur in dogs kept outdoors in the temperate climate of Japan. In summer dogs, heat production was significantly higher in cool environments, while it was significantly lower in thermoneutral and warm environments. Although the dogs used in this study were rather poorly insulated animals, an improvement in insulation was suggested in winter. Not only the improved insulation of the fur coat but also the increased metabolic rate at TNT seemed to shift the critical temperature to the lower level by approximately 4°C. Food intake increased in winter. The significantly increased heat production in the winter dogs at T_w above the lower critical temperature seemed to be attributable to the elevated resting metabolic rate.

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